Category Theory Leinster - Solutions Manual

Mingruifu Lin

September 2026

Contents

1 Categories, functors and natural Transformations **5**
1.1 Categories 5

Chapter 1

Categories, functors and natural Transformations

1.1 Categories

Exercise 1.1.12

I'm not someone with much ideas.

Exercise 1.1.13

Suppose both g_1 and g_2 are inverses of f. First, we have

$$g_1 f = 1_A$$

Multiply both sides by g_2

$$g_1 f g_2 = 1_A g_2$$

Since $f g_2 = 1_B$, we have

$$g_1 1_B = 1_A g_2$$

Note that the difference in identity makes sense (g_1 requires type B, but g_2 produces type A). Anyways, applying identity, we end up with

$$g_1 = g_2$$

Exercise 1.1.14

Composition is defined

$$\mathscr{A}((A', B'), (A'', B'')) \times \mathscr{A}((A, B), (A', B')) \to \mathscr{A}((A, B), (A'', B''))$$

$$(g_A \times g_B) \times (f_A \times f_B) \mapsto (g_A \circ f_A) \times (g_B \circ f_B)$$

And identity is defined

$$1_{(A,B)} = 1_A \times 1_B$$

Exercise 1.1.15

I don't understand the question, nor do I know what's a homotopy class.